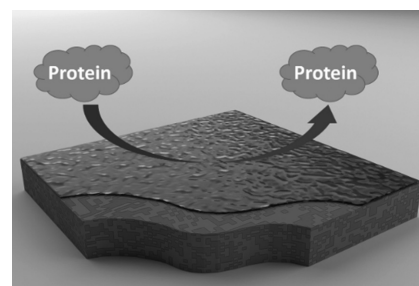


Improved Antifouling Properties of Polydimethylsiloxane Films via Formation of Polysiloxane/Polyzwitterion Interpenetrating Networks

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Nonspecific adsorption of proteins is a challenging problem for the development of biocompatible materials, as well as for antifouling and fouling-release coatings, for instance for the marine industry. The concept of preparing amphiphilic systems based on low surface energy hydrophobic materials via their hydrophilic modification is being widely pursued. This work describes a novel two-step route for the preparation of interpenetrating polymer networks of otherwise incompatible poly(dimethylsiloxane) and zwitterionic polymers. Changes in surface hydrophilicity as well as surface charge at different pH values are investigated. Characterization using atomic force microscopy provides thorough insight into surface changes upon hydrophilic modification. Protein fouling of the materials is assessed using fibrinogen as a model protein.



1. Introduction

Owing to their unique material properties and chemical stability, polydimethylsiloxane (PDMS) based materials are widely used for various applications, ranging from microfluidic devices to contact lenses, implants or medical catheters.^[1–3] Low toxicity, biocompatibility, elastic properties, and high gas permeability make PDMS the material of choice for above-mentioned applications. PDMS-based materials have also found application as environmentally friendly coatings against marine biofouling.^[4] Due to

their low mechanical modulus and low surface energy they show superior performance as fouling release materials.^[5–7]

On the other hand, however, the high surface hydrophobicity of PDMS materials makes them vulnerable against nonspecific adsorption of hydrophobic contaminants.^[8,9] Fouling of medical implants with proteins, bacteria, or cells upon contact with body fluids can lead to undesirable reactions of the immune system.^[10,11] It has also been found that certain microorganisms are able to foul PDMS-based coatings on marine vessels, thus providing a primary fouling layer for further adhesion and growth of more sophisticated organisms, eventually leading to high overall fouling.^[12]

Various strategies, physical as well as chemical, have been applied in order to enhance the antifouling and fouling release performance of PDMS materials. Physical modification methods are mostly based on the oxidation of the material surface, thus providing increased hydrophilicity via formation of silanol groups.^[1] Nevertheless, after physical treatment recovery of surface

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hydrophobicity occurs in a relatively short time due to rearrangement of PDMS chain segments through the process also referred to as hydrophobic recovery.^[13]

Modification of PDMS with polyethyleneglycol (PEG) has been found to be a convenient way to increase hydrophilicity, while maintaining the electrically neutral character and biocompatibility of the material.^[14,15] However, relatively poor chemical stability of PEG in the presence of oxygen and transition metal ions, or through enzymatic processes, has incited further interest into more stable, nonfouling and biocompatible materials.^[16–19] In recent years the modification of hydrophobic PDMS using zwitterionic polymers has emerged as a promising method, which increases hydrophilicity of the material and imparts fouling resistant properties.^[20–24] Much like for PEG-based systems, it has been shown that the introduction of highly hydrophilic zwitterionic moieties leads to the formation of a bound water layer at the material surface, thus creating thermodynamically unfavorable conditions for the adsorption or adhesion of various fouling species.^[19,25] It has also been shown that surfaces that are resistant to protein fouling are resistant to adhesion of bacteria or mammalian cells as well.^[26] Generally, both, “grafting-to” and “grafting-from” approaches have been reported to be applicable for the attachment of hydrophilic moieties on the material surface with respective advantages and disadvantages for both methods.^[27–29]

For example, surface modification of flexible hydrophobic matrices with PDMS-*b*-polysulfobetaine block-copolymers has been reported. Anchoring of these block copolymers into a PDMS or polyurethane matrix imparts fouling resistance properties to the base material due to formation of highly hydrophilic zwitterionic layer with zero net charge.^[20,21,30] Surface modification of PDMS via zwitterionic polymers for the reduction of protein attachment has been reported by Goda et al. In their work, UV-induced surface graft copolymerization method was employed to immobilize poly(2-methacryloyloxyethyl phosphorylcholine) on PDMS substrate.^[31]

Using an alternative approach, Turner and Cheng have developed functional membranes via modification of PDMS films through monomer immersion and cross-linking polymerization method.^[32–34] The authors have reported carboxyl functionalization of PDMS via formation of interpenetrating polymer network (IPN) structures leading to stimuli responsive behavior of obtained materials. This method offers a convenient way for modification of PDMS materials through the formation of strong, mechanically interlocked networks. Consequently, the analogous approach has been successfully employed by other groups as well to impart various functionalities and properties to PDMS materials.^[35,36] However, functionalization of PDMS with polyzwitterions by such an approach has not been reported yet.

In this work we report the modification of the hydrophobic PDMS via a simple two-step procedure, where each step represents a facile and scalable process. Introduction of otherwise incompatible zwitterionic polymers into PDMS matrix is realized by first forming an IPN using the well-compatible monomer 2-dimethylaminoethyl methacrylate (DMAEMA) and subsequent N-alkylation of tertiary amino groups of DMAEMA units, leading to formation of a PDMS–polyzwitterion IPN. The method represents a convenient tool for the modification of hydrophobic PDMS materials in order to improve its resistance to protein adsorption.

2. Experimental Section

2.1. Materials

Sylgard 184 was purchased from Dow-Corning. DMAEMA (98%; Sigma-Aldrich), 1,3-propane sultone (98%; Sigma-Aldrich), β -propiolactone (98%; Acros Organics), ethyleneglycol dimethacrylate (EGDMA, 98%; Sigma-Aldrich), Irgacure 651 (2,2-dimethoxy-2-phenylacetophenone, BASF), and fibrinogen (bovine fibrinogen fraction I; MP Biomedicals, USA) were used as received. Tetrahydrofuran (THF, 99.5%; Bernd Kraft, Germany) was dried using 3 Å molecular sieve. Sodium chloride, potassium chloride, potassium dihydrogen phosphate, and sodium hydrogen phosphate for phosphate buffered saline were also purchased from Sigma-Aldrich. Milli-Q water was used for preparation of buffer solutions, as well as for other experimental procedures.

2.2. Sample Preparation

PDMS–polyzwitterionic IPN films were obtained in a two-step procedure. First, the IPN was prepared using the functional monomer (DMAEMA), the cross-linker monomer (EGDMA), and cross-linked PDMS as the matrix network. Subsequent polymer-analogous reactions on tertiary amino groups of DMAEMA units of the second network lead to the formation of PDMS–polyzwitterion IPN. Detailed procedures are described below.

PDMS elastomer films with an average thickness of 0.5 mm were prepared from commercially available two-component Sylgard 184 (cf. Section 2.1) via curing for 12 h at 80 °C. Obtained films were weighed and then swollen in the mixture of DMAEMA and cross-linker monomer EGDMA in the presence of UV initiator Irgacure 651 for 3 h. 1 wt% EGDMA (relative to DMAEMA) and 1 wt% of Irgacure 651 (relative to total mixture—monomer and cross-linker) were used. Polymerization was initiated by UV irradiation leading to formation of interpenetrating networks of PDMS and PDMAEMA. UV-induced polymerization was conducted in UVACUBE 2000 (Honle UV Technology, Germany) equipped with UV-F lamp (emission mostly in UV-A and B regions) with irradiation intensity $\approx 45 \text{ mW cm}^{-2}$ for 40 min. During the polymerization, the samples were immersed into DMAEMA monomer to avoid leaching and evaporation of the monomer from swollen PDMS film surfaces. Afterward samples

were washed multiple times with Milli-Q water and dried in a vacuum oven for at least 24 h at room temperature. For further analysis at this stage, the samples were washed multiple times with THF to release trapped monomer and loosely cross-linked polymer. This additional treatment is required, since release of such material occurs during the subsequent polymer-analogous reaction (cf. below). The degree of functionalization with cross-linked PDMAEMA and the degree of swelling of the PDMS–PDMAEMA in water were determined gravimetrically.

Polymer-analogous alkylation of the PDMS–PDMAEMA IPN was conducted as follows. Dried PDMS–PDMAEMA samples were placed in dry THF and an excess (double) amount of 1,3-propane sultone or β -propiolactone (with respect to moles of repeating units of PDMAEMA in the sample) was added. Samples were left to react for 72 h with mild stirring on a magnetic stirrer, leading to transformation of tertiary amino groups of PDMAEMA into sulfobetaine and carbobetaine groups and thus yielding poly(sulfobetaine methacrylate) (PSBMA) and poly(carbobetaine methacrylate) (PCBMA), respectively.^[37] Afterward samples were washed multiple times using THF and water.

2.3. Characterizations

Swelling degree (SD) of unmodified and modified samples in various solvents was investigated gravimetrically and calculated as follows: $SD = [(W_s - W_d)/W_d] \times 100$, where W_s and W_d are the weights of swollen and dry samples, respectively. The same method was used to determine the water uptake of modified samples.

Infrared spectra (IR) were recorded in attenuated total reflection (ATR) mode using Varian 3100 ATR-FTIR Excalibur Series equipped with a horizontal ATR unit with a diamond/KRS-5 crystal and an angle of incidence of 45°.

Contact angle values were measured using an optical measurement system (OCA 15, Dataphysics, Germany). In order to avoid potential errors caused by swelling and surface rearrangement of dry samples in water, the contact angle measurements were conducted in captive bubble mode. Prior to measurements the sample films were equilibrated for 24 h in water. Zeta potentials of the samples were measured using SurPASS electrokinetic analyzer (Anton-Paar, Austria) in standard electrolyte, 1×10^{-3} M KCl solution; more details on procedure and calculations can be found in previous publications.^[30,38]

Atomic force microscopy (AFM) measurements were conducted for topographic imaging and nanoindentation tests using a Dimension ICON instrument (Bruker, Germany). For AFM studies, small pieces of unmodified PDMS and PSBMA-modified PDMS were immobilized in a Petri dish (working as a fluid cell) and filled with Milli-Q water. To obtain topographic images, samples were measured in PeakForce mode in water using a silicon nitride cantilever with a nominal spring constant of 0.7 N m^{-1} . The same cantilever was also used to measure force–distance curves.

In order to evaluate fibrinogen adsorption, the discs of the above-prepared modified and unmodified PDMS films with diameter of 6.5 mm were first equilibrated in PBS buffer and then immersed in 400 μL protein solution with Fg concentration $10 \mu\text{g mL}^{-1}$ in phosphate buffered saline (PBS) buffer for 3 h.

Afterward the solutions were analyzed using MicroBCA protein assay kit (Thermo Fischer Scientific). UV–vis microplate reader μQuant (BioTek Instruments) was used for quantification.

3. Results and Discussion

The sequential modification procedure reported here enables introduction of highly hydrophilic zwitterionic polymers into otherwise incompatible highly hydrophobic PDMS films with the formation of a chemically interlocked structure. The target is achieved via introduction of the tertiary amino groups into PDMS matrix in a first step by formation of interpenetrating polymeric network structure via a monomer immersion method, which is easily performed due to high compatibility of PDMS with respective monomer (DMAEMA) and subsequent photoinitiated cross-linking polymerization. Upon swelling the PDMS films in monomer/cross-linker mixture, average swelling degree of $80.5 \pm 1.5\%$ was observed. Upon UV polymerization and multiple washing with water $86.5 \pm 2.2\%$ weight increase was observed compared to unmodified PDMS samples. After extraction of the films with THF (the solvent used for the second step, the polymer-analogous reaction), the PDMS–PDMAEMA IPN showed a weight gain (degree of modification) of $39.6 \pm 0.7 \text{ wt}\%$ compared to the precursor PDMS. Hence, about 50% of the monomer reaction mixture had been converted to the second network; the extractable material consisted presumably of linear or weakly cross-linked PDMAEMA, oligomers, and monomer. The low efficiency of extraction of that material by water can be related to the much lower degree of swelling of PDMA–PDMAEMA IPN in water ($\approx 52\%$) compared to THF ($\approx 168\%$; cf. below).

In the second step, the tertiary amino groups are converted into zwitterionic units through a straightforward alkylation process using 1,3-propane sultone and β -propiolactone. Both compounds show high reactivity toward tertiary amines and it had been reported that high conversions can be achieved within several hours in homogeneous reactions even at room temperature.^[30,37,38] Considering the diffusion-limited reaction rate in the here-described interpenetrating polymer networks, the samples were allowed to react for prolonged times (72 h; cf. Section 2.2). Increased reaction time in combination with strong swelling of PDMS–PDMAEMA in reaction solvent THF (swelling degree $167.5 \pm 3.5\%$) could potentially counteract steric hindrance to achieve effective penetration of reactants into the swollen bulk material as well. Gravimetric analyses revealed the complete sulfobetainization (within the range error, i.e., 5%), while degree of conversion to carbobetaines was found to be only $55.2 \pm 4.9\%$. The reason for this difference is not clear, and considering the used reaction conditions (cf. above) there

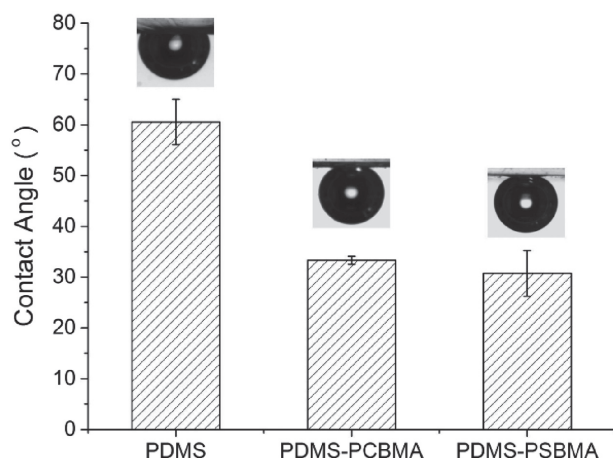


Figure 1. Water contact angle values, obtained in captive air bubble mode, for unmodified and modified PDMS films.

is no obvious way to increase degree of conversion in the case of PDMS–PCBMA. Stepwise conversion was also followed by ATR-FTIR spectroscopy, penetration depth of which is limited to few micrometers from the sample surface. Respective spectra are represented in Supporting Information (Figure S1). The formation of IPN structures with PDMS was confirmed, but no more details regarding the different specific structures could be obtained with that method.

Introduction of hydrophilic polymers into PDMS films naturally leads to the increased surface hydrophilicity. Contact angle measurements in captive bubble mode after equilibrating the films for 24 h enable quantitative assessment of surface wetting properties, and results are provided in Figure 1. As shown, strong decrease of contact angle values is observed upon formation of IPN structures of PDMS and zwitterionic polymers, reducing from initial $60.5^\circ \pm 4.5^\circ$ for unmodified PDMS, down to $33.3^\circ \pm 0.7^\circ$ and $30.7^\circ \pm 4.5^\circ$ for PCBMA and PSBMA-modified films, respectively. Contact angle value of the intermediate product PDMS–PDMAEMA was found to be $36.3^\circ \pm 2.5^\circ$, i.e., significantly higher than for the polyzwitterionic derivative of the IPN. Because of the very smooth surfaces (see AFM data below) normalization of the contact angle values by surface roughness did not have influence on the results. Increased hydrophilicity also leads to high water uptake by modified samples, i.e., up to $111.9 \pm 11.1\%$ and $64.7 \pm 2.9\%$ for PCBMA and PSBMA-modified PDMS films, respectively. Water uptake of PDMS–PDMAEMA IPN films was $52.5 \pm 4.2\%$ and thus noticeably lower compared to polyzwitterionic-modified films. Relatively large difference in water uptake values of the two polyzwitterionic-modified films can possibly be assigned to the different degrees of modification or different compatibilities of PSBMA and PCBMA with PDMS matrix. Lower conversion, observed for carbobetaine-containing films, could also

potentially increase the water accessibility of protonated tertiary amino groups in the bulk material, which show stronger repulsion between positively charged repeat units and corresponding higher hydration. Moreover, different hydration behavior of the zwitterionic betaines has been studied and reported in the literature.^[39] Molecular simulation studies have shown that negatively charged groups of carbobetaines and sulfobetaines exhibit noticeably different hydration because of different degree of ion pairing between cation and anion in the side group. Water molecules around the negatively charged group of carbobetaines have sharper spatial distribution, preferential dipole orientation, and longer residence time, and carbobetaines show larger water uptake than sulfobetaines in equilibrium.

Zeta potentials of unmodified and modified PDMS films were measured as a function of the pH value. Unmodified PDMS films exhibit negative zeta potential at pH values above 5 (Figure 2). This can be explained by the adsorption of electrolyte ions on the hydrophobic polymer surface, which is more effective for anions than for cations (cf.^[38]). In comparison to unmodified films the zeta potential values of sulfobetaine and carbobetaine-modified films have been shifted toward neutral and slightly positive values, which can be explained by the presence of the surface layer on the modified PDMS samples, consisting of cross-linked zwitterionic polymer. It should be noted that in physiologically relevant pH range (around pH 7.4) zeta potential values are only very slightly positive for modified samples, thus reducing the electrostatic attraction of foulants to a minimum.

Considering the conclusions from the surface analyses, the lower degree of conversion for carbobetaines compared to sulfobetaine seems to be relevant only for the bulk of the IPN films; the surface of both IPN types

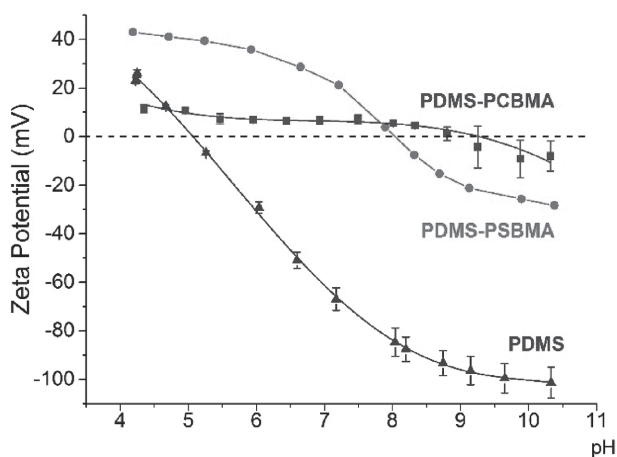


Figure 2. Zeta potential as a function of the pH for modified and unmodified PDMS films.

showed similar properties that are dominated by the polyzwitterionic character of the second network.

AFM investigation of the unmodified and PSBMA-modified films in aqueous conditions reveals no significant changes of surface topographies upon modification (Figure S2, Supporting Information). There may be indications for a slightly rougher surface along with some nodular features for the PSBMA-PDMS compared to PDMS films; however, it should be considered that the topography analysis is also influenced by the “soft” nature of both materials. An overview on additional data for relative surface roughness obtained by scanning different sample areas is presented in Table S1 (Supporting Information). Relative surface roughness, i.e., the ratio between actual and geometric (projected) area, was 1 within the range of error for both materials. Obtained films exhibit smooth surfaces and normalization of the contact angle values by surface roughness does not influence the results noticeably (see the Supporting Information).

Hence, for more thorough characterization, force–distance curves were obtained by indenting the films with the AFM tip and observing the resulting deflection of the cantilever, and these curves were then converted into force–separation curves.^[40] In order to remove offset and tilt, curves have been baseline corrected via software built-in function. The initial contact point (tip-sample separation = 0) is set to the position where the maximum pull-on adhesion in the approach curve is reached. This makes indentation depths of the different samples directly comparable.

The first force–separation curve displayed in Figure 3a was obtained on the silicon wafer and was used for estimating the cantilever deflection sensitivity. No indentation is observable in the curve, as expected for rigid materials like silicon.

The curve for the unmodified PDMS displayed in Figure 3b exhibits the typical behavior of elastic materials wherein the loading force yields to an appreciable indentation depth, around 50 nm indentation at 10 nN loading force. Compared to this, the indentation depth for the PDMS–PSBMA sample at similar loading force (here 8 nN vs. 10 nN for unmodified PDMS) is nearly ten times larger indicating a very soft surface that is expected for a swollen hydrogel (Figure 3c).

Fouling resistance behavior of the films was assessed using fibrinogen as model protein. Fibrinogen is the major protein mediating platelet adhesion to hydrophobic polymeric materials, and it has an isoelectric point (IEP) of 5.8.^[41] Besides, it adsorbs to larger extent onto hydrophobic surfaces in comparison to bovine serum albumin (BSA).^[41,42] In fact, BSA was found not suitable for this study due to its lower adsorption values and resulting limitations of the MicroBCA protein assay-based method used (cf. Section 2.3). Upon incubation

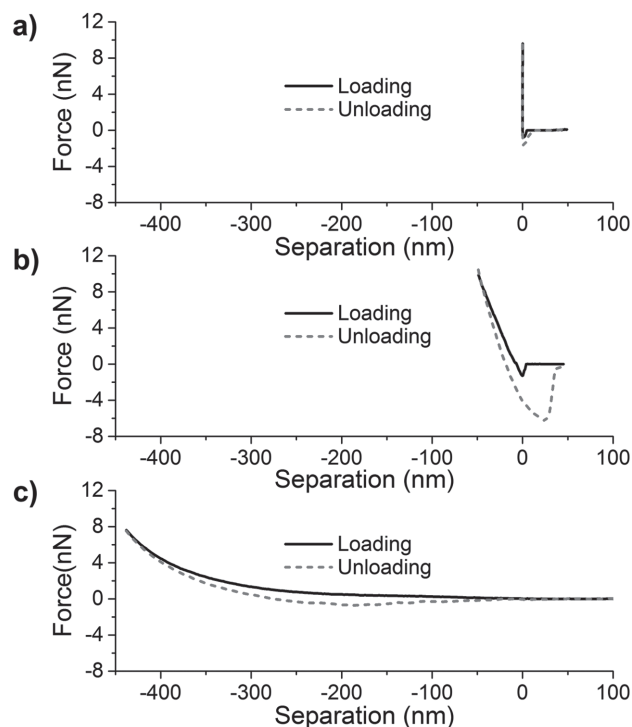


Figure 3. Force–separation curves of AFM tip and surface to be analyzed: a) silicon wafer, b) unmodified PDMS film, and c) PDMS–PSBMA film equilibrated in Milli-Q water.

with fibrinogen solution, significant reduction of protein adsorption is observed due to modification of the samples with zwitterionic polymers. Amounts of adsorbed Fg per unit area are $3.34 \pm 0.55 \mu\text{g cm}^{-2}$, $1.13 \pm 0.11 \mu\text{g cm}^{-2}$, and $0.84 \pm 0.01 \mu\text{g cm}^{-2}$ for unmodified, PSBMA, and PCBMA-modified PDMS films, respectively. The results are graphically represented in Figure 4 in a normalized form. Even though the employed experimental technique does not provide the possibility to distinguish between reversibly and irreversibly adsorbed Fg, absolute values of adsorbed

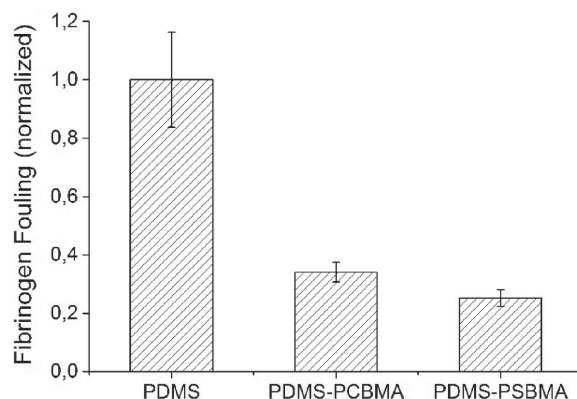


Figure 4. Fouling of unmodified and modified PDMS films with bovine fibrinogen as amount of adsorbed protein per surface area (fouling of unmodified PDMS is normalized to 1).

protein are in a similar range as reported in the literature.^[43] Also, it cannot be excluded that some protein may have penetrated into the swollen IPN so that obtained data would be an overestimation of adsorptive fouling. That uptake of biomacromolecules into water swollen PDMS-based IPN is possible had been demonstrated for carboxyl-functional materials and dextran.^[44] Considering the IEP of Fg, the slightly positive excess charge of the polyzwitterion-PDMS IPN of this study (cf. Figure 4), and the high swelling in water, this scenario is possible.

Nevertheless, a clear correlation between protein adsorption and surface hydrophilicity can be observed (Figures 1 and 4). These observations further support the model suggested by Whitesides and co-workers^[25] and Grunze and co-workers.^[25,45] According to this model, immobilization of water molecules due to strong hydration of highly hydrophilic groups is responsible for deterring the protein adhesion to the surface, with a relatively minor role of steric repulsion. Strong hydration of the polyzwitterion-modified surfaces is further supported by AFM investigations of the sample in aqueous conditions. Soft, gel-like character of the surface is a likely result of high fraction of hydrophilic polymer in the interpenetrating network structure and the resulting high water uptake.

4. Conclusions

In summary, we have reported the facile preparation by two steps, interpenetration polymeric network formation and polymer-analogous functionalization, and the fouling resistant behavior of modified PDMS-based materials. Changes observed upon polyzwitterionic modification and the final characteristics of the modified materials indicate that the reduction of protein adhesion is in correlation with increased surface hydrophilicity, which strongly reduces the hydrophobic interactions while minimizing the electrostatic interactions between fouling protein and the substrate. Presence of the soft hydrated layer on the materials surface, observed by AFM measurements, further supports this hypothesis. The reported method extends significantly the available toolbox used for the incorporation of zwitterionic moieties into hydrophobic polymeric matrices and can potentially be scalable, increasing the applicability of the method. The hydrophobic/hydrophilic domain character of the materials on the surface and in the bulk as function of the boundary conditions, as well as the resulting partitioning of solutes into the amphiphilic IPNs, will be studied in more detail. Controlled swelling of the material surface upon immersion into monomer/cross-linker mixture can potentially lead to optimized mechanical stability and

other properties and thus further improve applicability of the established systems.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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